

House Price Prediction — Summary of Findings

Dataset: Housing Prices Dataset (Kaggle) — 545 properties, 12 features.
Goal: Predict house price and identify the strongest price drivers.

Model Performance

Metric	Linear Regression	Random Forest
MAE (Rs)	970,043	1,019,528
RMSE (Rs)	1,324,507	1,400,219
R ² Score	0.6529 (65.3%)	0.6121 (61.2%)

Key Findings & Recommendation

The three features that influence house price the most are area, number of bathrooms, and air conditioning, with plot area standing out as the strongest driver of price. The Random Forest model explains about 61% of the variation in house prices ($R^2 = 0.61$) and has an average error of roughly Rs 1,019,528 per house, while the Linear Regression model explains about 65% of the variation ($R^2 = 0.653$). These models provide useful ballpark estimates but should not be treated as exact valuations. Bedroom count contributes less than expected once plot area is considered, whereas air conditioning has a stronger positive impact on price. Based on these findings, real estate businesses should prioritize plot area and bathroom count when pricing properties, while highlighting air conditioning and preferred locations as value-adding features.

Top 3 Price Drivers

1. Area
2. Bathrooms
3. Air Conditioning